

Test the hypotheses that the yearly amount of bananas consumed is unrelated to the yearly amount of

- (a) Apples consumed
- (b) Oranges consumed
- (c) Test the hypothesis that the yearly per capita amount of oranges consumed is unrelated to the yearly amount of apples consumed.

12.6 REGRESSION TO THE MEAN

The term *regression* was originally employed by Francis Galton while describing the laws of inheritance. Galton believed that these laws caused population extremes to “regress towards the mean.” By this he meant that children of individuals having extreme values of a certain characteristic would tend to have less extreme values of this characteristic than their parents.

If we assume a linear regression relationship between the characteristic of the offspring Y and that of the parent x , then a regression to the mean will occur when the regression parameter β is between 0 and 1. That is, if

$$E[Y] = \alpha + \beta x$$

and $0 < \beta < 1$, then $E[Y]$ will be smaller than x when x is large and will be greater than x when x is small. That this statement is true can be easily checked either algebraically or by plotting the two straight lines

$$y = \alpha + \beta x$$

and

$$y = x$$

A plot indicates that when $0 < \beta < 1$, the line $y = \alpha + \beta x$ is above the line $y = x$ for small values of x and is below it for large values of x . Such a plot is given in Fig. 12.4.

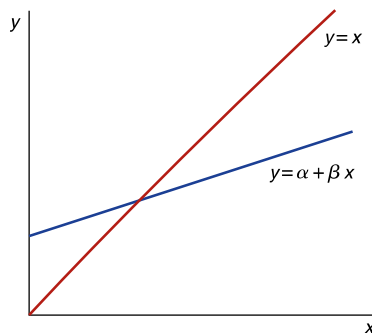


FIGURE 12.4

Regression to the mean occurs when $0 < \beta < 1$. For x small, $\alpha + \beta x > x$; for x large, $\alpha + \beta x < x$.

■ Example 12.5

To illustrate Galton's thesis of regression to the mean, the British statistician Karl Pearson plotted the heights of 10 randomly chosen sons versus those of their fathers. The resulting data (in inches) were as follows.

Father's height	Son's height	Father's height	Son's height
60	63.6	67	67.1
62	65.2	68	67.4
64	66	70	68.3
65	65.5	72	70.1
66	66.9	74	70

A scatter diagram representing these data is presented in Fig. 12.5.

Note that whereas the data appear to indicate that taller fathers tend to have taller sons, they also appear to indicate that the sons of fathers who are either extremely short or extremely tall tend to be more "average" than their fathers; that is, there is a *regression toward the mean*.

We will determine whether the preceding data are strong enough to prove that there is a regression toward the mean by taking this statement as the alternative hypothesis. That is, we use the given data to test

$$H_0: \beta \geq 1 \quad \text{against} \quad H_1: \beta < 1$$

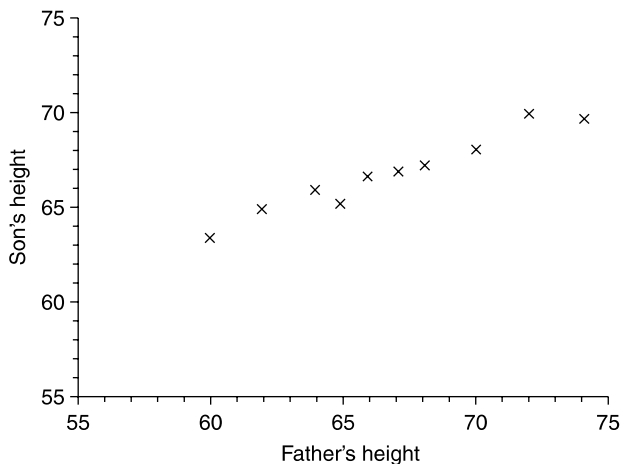


FIGURE 12.5

Scatter diagram of son's height versus father's height.

Now, this test is equivalent to a test of

$$H_0: \beta = 1 \quad \text{against} \quad H_1: \beta < 1$$

and will be based on the fact that

$$\sqrt{\frac{(n-2)S_{xx}}{SS_R}}(\hat{\beta} - \beta)$$

has a t distribution with $n - 2$ degrees of freedom.

Hence, when $\beta = 1$, the test statistic

$$TS = \sqrt{\frac{8S_{xx}}{SS_R}}(\hat{\beta} - 1)$$

has a t distribution with 8 degrees of freedom. The significance-level- α test will be to reject H_0 when the value of TS is sufficiently small (since this will occur when $\hat{\beta}$, the estimator of β , is sufficiently smaller than 1). Specifically, the test is to

$$\text{Reject } H_0 \quad \text{if } TS \leq -t_{8,\alpha}$$

$$\text{Not reject } H_0 \quad \text{Otherwise}$$

To determine the value of the test statistic TS, we run Program 12-1 and obtain the following:

The least-squares estimators are as follows

$$A = 35.97757$$

$$B = 0.4645573$$

The estimated regression line is

$$Y = 35.97757 + 0.4645573x$$

$$S(x,Y) = 79.71875$$

$$S(x,x) = 171.6016$$

$$S(Y,Y) = 38.53125$$

$$SS_R = 1.497325$$

$$\text{The square root of } (n-2)S(x, x)/SS_R \text{ is } 30.27942$$

From the preceding we see that

$$TS = 30.2794(0.4646 - 1) = -16.21$$

Since $t_{8,0.01} = 2.896$, we see that

$$TS < -t_{8,0.01}$$

and so the null hypothesis that $\beta \geq 1$ is rejected at the 1 percent level of significance. In fact the p value is

$$p \text{ value} = P\{T_8 \leq -16.213\} \approx 0$$

and so the null hypothesis that $\beta \geq 1$ is rejected at almost any significance level, thus establishing a regression toward the mean. ■

A modern biological explanation for the phenomenon of regression to the mean would roughly go along the lines of noting that since an offspring obtains a random selection of one-half of each parent's genes, it follows that the offspring of a very tall parent would, by chance, tend to have fewer "tall" genes than its parent.

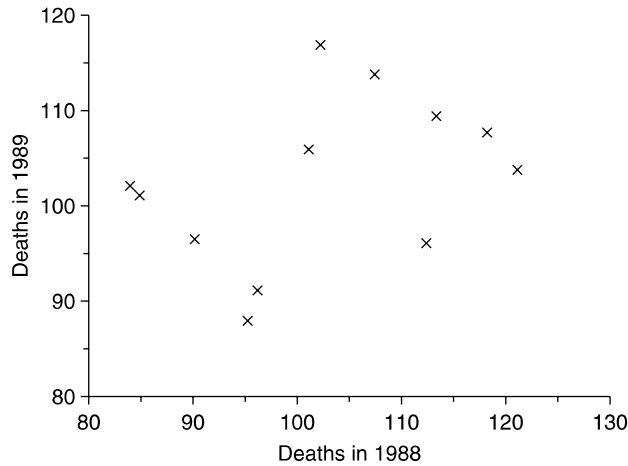
While the most important applications of the phenomenon of regression to the mean concern the relationship between the biological characteristics of an offspring and those of its parents, this phenomenon also arises in situations where we have two sets of data referring to the same variables. We illustrate it in Example 12.6.

■ Example 12.6

The following data relate the number of motor vehicle deaths occurring in 12 counties in the northwestern United States in the years 1988 and 1989.

County	Deaths in 1988	Deaths in 1989
1	121	104
2	96	91
3	85	101
4	113	110
5	102	117
6	118	108
7	90	96
8	84	102
9	107	114
10	112	96
11	95	88
12	101	106

The scatter diagram for this data set appears in Fig. 12.6. A glance at Fig. 12.6 indicates that in 1989 there was, for the most part, a reduction in the number

**FIGURE 12.6**

Scatter diagram of 1989 deaths versus 1988 deaths.

of deaths in those counties that had a large number of motor vehicle deaths in 1988. Similarly, there appears to have been an increase in those counties that had a low value in 1988. Thus, we would expect that a regression to the mean is in effect. In fact, running Program 12-1 yields the estimated regression equation

$$\hat{y} = 74.589 + 0.276x$$

which shows that the estimated value of β indeed appears to be less than 1.

One must be careful when considering the reason behind the phenomenon of regression to the mean in the preceding data. For instance, it might be natural to suppose that those counties that had a large number of deaths caused by motor vehicles in 1988 would have made a large effort—perhaps by improving the safety of their roads or by making people more aware of the potential dangers of unsafe driving—to reduce this number. In addition, we might suppose that those counties that had the fewest number of deaths in 1988 might have “rested on their laurels” and not made much of an effort to further improve their numbers—and as a result had an increase in the number of casualties the following year.

While the foregoing suppositions might be correct, it is important to realize that a regression to the mean would probably have occurred even if none of the counties had done anything out of the ordinary. Indeed, it could very well be the case that those counties having large numbers of casualties in 1988 were just very unlucky in that year, and thus a decrease in the next year was just a return to a more normal result for them. (For an analogy, if 9 heads result when

10 fair coins are flipped, then it is quite likely that another flip of these 10 coins will result in fewer than 9 heads.) Similarly, those counties having few deaths in 1988 might have been “lucky” that year, and a more normal result in 1989 would thus lead to an increase.

The mistaken belief that regression to the mean is due to some outside influence, when it is in reality just due to “chance,” is heard frequently enough that it is often referred to as the *regression fallacy*. ■

Regression to the mean plays a key role in the explanation of why so many communal biological data sets from a homogeneous population tend to have “normal curve” histograms. For instance, if one plotted the heights of all senior girls in a specified high school, then it is a good bet that the resulting histogram would strongly resemble the bell-shaped normal curve. One possible explanation for this combines the central limit theorem, regression to the mean, and the passing of many generations. We now sketch it.

*12.6.1 Why Biological Data Sets Are Often Normally Distributed

We will present this argument in the context of considering the heights of females in a population. We will follow this population of females over many generations. Suppose that there are initially k women, whom we will refer to as the *initial generation*, and that their heights are $x_1, \dots, x_k$. These k values are considered to be totally arbitrary. Let d denote the largest minus the smallest of these values. For instance, if

$$k = 3 \quad x_1 = 60 \quad x_2 = 58 \quad x_3 = 66$$

then $d = 66 - 58 = 8$.

If Y denotes the height of a female child of a woman whose height is x , then we will assume the linear regression model

$$Y = \alpha + \beta x + e$$

In this model we will make the usual assumption that e is an error random variable that is normally distributed with mean 0 and variance σ^2 . However, whereas this assumption is often made without any real attempt at justification, it seems quite reasonable in this application because of the central limit theorem. That is, the height of a daughter of a woman of height x can be thought of as being composed of the sum of a large number of approximately independent random variables that relate, among other things, to the random set of genes that she receives as well as to environmental factors. Hence, by the central limit theorem, her height should be approximately normally distributed. We will also assume that regression to the mean is in effect, that is, that $0 < \beta < 1$.

Thus, the heights of the daughters of the k women of the initial population are all normally distributed. However, it is important to note that their mean heights are all different. For instance, a daughter of the woman of height x_1 will have a normally distributed height with mean value $\alpha + \beta x_1$, whereas the daughter of the woman of height x_2 will have a different mean height, namely, $\alpha + \beta x_2$. Thus, the heights of all the daughters do *not* come from the *same* normal distribution, and for that reason a plot of all their heights would not follow the normal curve.

However, if we now consider the difference between the largest and the smallest *mean* height of all the daughters of the initial set of women, then it is not difficult to show that

$$\text{Difference} \leq \beta d$$

(If each woman of the initial set of women had at least one daughter, then this inequality would be an equality.) If we now consider the daughters of these daughters, then it can be shown that their heights will be normally distributed with differing means and a common variance. The difference between the largest and the smallest mean height of these second-generation daughters can be shown to satisfy

$$\text{Difference} \leq \beta^2 d$$

Indeed, if we suppose that more and more generations have passed and we consider the women of the n th generation after the initial population, then it can be shown that the heights of the women in this generation are normally distributed with the same variance and with mean values that, while differing, are such that the difference between the largest and smallest of them satisfies

$$\text{Difference} \leq \beta^n d$$

Now, since $0 < \beta < 1$, it follows that as n grows larger, $\beta^n d$ gets closer and closer to 0. Thus, after a large enough number of generations have passed, all the women in the population will have normally distributed heights with approximately the same mean and with a common variance. That is, after many generations have passed, the heights of the women will come from approximately the same normal population, and thus at this point a plot of these heights will approximately follow the bell-shaped normal curve.

PROBLEMS

1. The following data come from an experiment performed by Francis Galton. The data relate the diameter of an offspring seed to that of its parent seed in the case of a self-fertilized seed.

Diameter of parent seed	Diameter of offspring seed
15	15.3
16	16.4
17	15.5
18	16.2
19	16.0
20	17.4
21	17.5

- (a) Estimate the regression parameters.
 (b) Does there appear to be a regression to the mean?
2. In Example 12.6 it was shown that the estimated value of β is less than 1. Using the data of this example, test the hypothesis

$$H_0: \beta \geq 1 \quad \text{against} \quad H_1: \beta < 1$$

Would H_0 be rejected at the 5 percent level of significance?

3. Would you be surprised if the following data sets exhibited a regression to the mean? Would you expect them to exhibit this phenomenon? Explain your answers.
- (a) You go to 10 different restaurants that you know nothing about in advance. You eat a meal in each one and give a numerical ranking—anywhere from 0 to 100—to the quality of the meal. You then return to each of these restaurants and again give a ranking to the meal. The data consist of the two scores of each of the 10 restaurants.
- (b) At the beginning of an hour, 12 individuals check their pulse rates to determine the number of heartbeats per minute. Call these the x values. After 1 hour, they repeat this to obtain the y values.
- (c) The set considers paired data concerning different mutual funds. For each mutual fund, the x variable is the 1995 ranking of the fund, and the corresponding y variable is the 1996 ranking.
- (d) The data consist of the paired scores of 20 first-year preschoolers, with the first value in the pair being the student's test score on an IQ examination given to all entering students and the second value being the same student's score on an IQ test given at the end of the first month in school.
4. Test

$$H_0: \beta = 1 \quad \text{against} \quad H_1: \beta < 1$$

for the following set of data. Use the 5 percent level of significance.

x	y
24	27
21	24
26	20
17	22
15	21
24	20
23	17

5. The following are the average 2000 and 2002 math SAT scores in a sample of states.

State	2000	2002
Arizona	523	523
California	518	517
Indiana	501	503
Missouri	577	580
Florida	486	473
Oregon	527	528
Virginia	500	506
Wisconsin	597	599
Texas	500	500
Vermont	508	510

- (a) Find the estimated regression line.
 (b) Does it indicate a regression to the mean?
6. Figure 12.7 presents a histogram of the heights of 8585 men. How well does it appear to fit a normal curve?

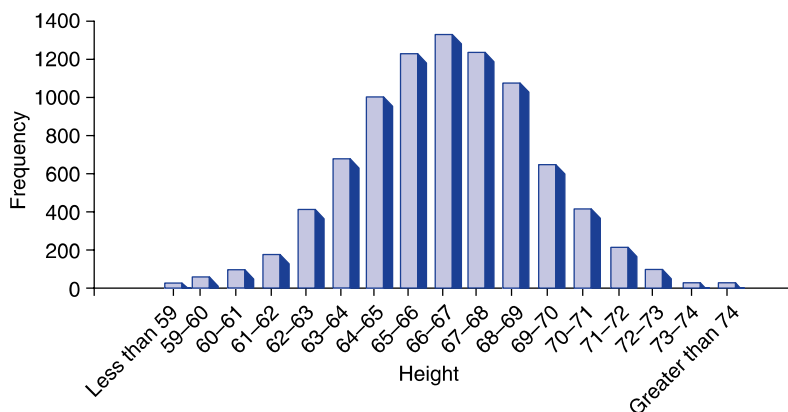


FIGURE 12.7

A histogram of heights.

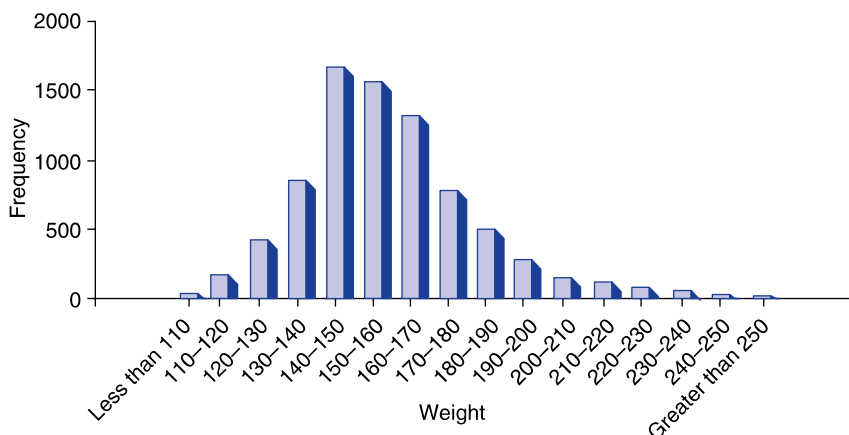


FIGURE 12.8
A histogram of weights.

7. Figure 12.8 presents a histogram of the weights of 7738 men. How well do these data fit a normal curve?

12.7 PREDICTION INTERVALS FOR FUTURE RESPONSES

Suppose, in the linear regression model, that input values x_i have led to the response values y_i , $i = 1, \dots, n$. The best prediction of the value of a new response at input x_0 is, of course, $\hat{\alpha} + \hat{\beta}x_0$. However, rather than give a single number as the predicted value, it is often more useful to be able to present an interval that you predict, with a certain degree of confidence, will contain the response value. Such a *prediction interval* is given by the following.

Prediction interval for a response at input value x_0 , based on the response values y_i at the input values x_i , $i = 1, \dots, n$:

With $100(1 - \gamma)$ degree confidence, the response Y at the input value x_0 will lie in the interval

$$\hat{\alpha} + \hat{\beta}x_0 \pm t_{n-2, \gamma/2} W$$

where $t_{n-2, \gamma/2}$ is the $100(1 - \gamma/2)$ th percentile of the t distribution with $n - 2$ degrees of freedom, and

$$W = \sqrt{\left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right] \frac{SS_R}{n-2}}$$